How do we find a basis for the column space of a matrix?

Answer:

To find a basis for the column space of a matrix, you can follow these steps:

1. **Perform Gaussian Elimination**: Start by performing Gaussian elimination on your matrix **A** to bring it to its row echelon form (REF). This process involves using row operations to simplify the matrix. Typically, you aim to introduce zeros below each pivot (leading non-zero entry in each row).
2. **Identify Pivot Columns**: In the REF of the matrix, identify the columns that contain the pivots. These pivot columns are crucial because they indicate which of the original columns of the matrix are linearly independent.
3. **Select Original Columns**: Go back to the original matrix **A** and select the columns corresponding to the pivot columns identified in the REF. These columns form a basis for the column space of the matrix.
4. **Form the Basis**: The columns selected in the previous step are linearly independent and span the column space of the matrix, thus forming a basis.

Here's an example for clarity:

Suppose you have a matrix **A**

$$\boldsymbol{A} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{pmatrix}$$

After performing Gaussian elimination, you might get the REF as follows:

$$\text{REF of } \boldsymbol{A} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

In this REF, the pivot columns are the first and second columns. Thus, the corresponding columns from the original matrix **A** form a basis for the column space:

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

These vectors are linearly independent and span the column space of **A**.

This method ensures that you have the minimal set of vectors necessary to describe the span of the column space.